

# A Feynman–Wigner–Style Diagnostic for the Efficacy of Conformal Prediction via Signed de Finetti Representations

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## Abstract

Conformal prediction builds prediction sets that cover the truth at a rate you choose, finite-sample and distribution-free, assuming only exchangeable data. That guarantee is marginal. de Finetti’s theorem describes the exchangeability it rests on, and in the finite form the mixing measure may be signed (?). A short lemma decomposes the slope of conformal’s calibration-conditional coverage into a non-negative threshold term (the classical Beta-law fan) and a term carrying the sign of the cross-sample dependence. When the test score rises stochastically with the calibration width, as for a scalar latent with log-concave noise, conformal is conditionally adaptive. When it falls, as for scores defined relative to the batch (de-measured, ranked, compositional), it is anti-adaptive. The sign of the finite de Finetti measure is related to this but does not determine it: a positive mixture need not be associated, and a law that needs a signed representation can be positively correlated. The marginal guarantee is the same either way. Only what it hides changes.

## 1 Setup

Split conformal returns a set  $C(X)$  with  $\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$  for any distribution and any sample size, provided the data are exchangeable. That marginal guarantee is the whole appeal, and it is not in dispute here. The question is what it says about the case in front of you.

Scores  $S_1, \dots, S_n$  (calibration) and  $S_{n+1}$  (test) are jointly exchangeable with a continuous joint law (no ties). The conformal width is the order statistic  $\hat{q} = S_{(k)}$  of the calibration scores,  $k = \lceil (n+1)(1-\alpha) \rceil$ , and a test point is covered when  $S_{n+1} \leq \hat{q}$ .

Two kinds of dependence live in any such problem, and they should not be confused.

- Within a row: inside a single  $(x, y)$ , does the residual’s shape depend on  $x$ ? That is the subject of the companion paper (?), where the log-score regret of a single-shape conformal system equals the mutual information  $I(R; X)$ . It is a property of one observation’s law, present even when the rows are i.i.d.
- Across rows: how do distinct observations relate to one another, whether independent, sharing a latent cause, or mutually constrained? That is the subject of de Finetti’s theorem, and of this note.

The two axes are orthogonal. This note is entirely about the second.

## 2 Finite de Finetti and the sign of the mixture

de Finetti’s theorem pins down what infinite exchangeability is.

**Theorem 1** (de Finetti; ??). *Let  $X_1, X_2, \dots$  be an infinitely exchangeable sequence in a Polish space. Then there is a probability measure  $\pi$  on the space of distributions such that, for every  $n$ , the law of  $(X_1, \dots, X_n)$  equals  $\int P^{\otimes n} d\pi(P)$ . Equivalently, conditional on a random measure  $\Theta \sim \pi$ , the  $X_i$  are i.i.d. with law  $\Theta$ .*

The mixing measure  $\pi$  is a genuine prior, and that forces a sign. A positive mixture has  $\text{Cov}(S_i, S_j) = \text{Var}(\mathbb{E}[S_i | \Theta]) \geq 0$ , so de Finetti mixtures are non-negatively correlated. Finite exchangeable sequences are not so constrained. Exchangeability alone forces only  $\rho \geq -1/(n-1)$  (?), and the negative band is real—sampling without replacement sits at its floor.

The finite theory completes the picture. ? show a finite exchangeable sequence is approximately a positive i.i.d. mixture, within total variation  $O(k/n)$ . ? make it exact, at the price of signing.

**Theorem 2** (Kerns–Székely; ?). *Let  $(X_1, \dots, X_n)$  be exchangeable on a measurable space. Then there is a finite signed measure  $\mu$  of total mass 1 on the space of distributions with  $\text{law}(X_1, \dots, X_n) = \int P^{\otimes n} d\mu(P)$ . One may take  $\mu \geq 0$  (a genuine prior, recovering Theorem 1) if and only if the sequence extends to an infinitely exchangeable one (?).*

We picture this as the i.i.d. laws tracing a curve: positive mixtures are its convex hull, the extendable laws, and signed mixtures its affine span, everything. A representation that must be signed marks non-extendability, a finite exchangeable sequence that cannot be continued to an infinite one. Signed representations are also not unique.

We frame the signed measure as a negative probability, the kind ? used partway through a calculation, or ?’s phase-space distribution, which dips below zero while every observable marginal stays a genuine probability (the idea reached statistics through ?). You cannot sample a parameter from it. That is why the signed corner gives a diagnosis, not a model: recovering a latent means fitting a real one, not reading the quasi-probability off.

Three properties need to be kept apart.

1. Extendability, equivalently a positive representation (a genuine prior).
2. The sign of the pairwise correlation.
3. Association or negative association (??), the monotone dependence that the slope in Section 4 needs.

Property 1 implies non-negative pairwise correlation, so negative correlation implies a signed representation. Neither converse holds, and property 1 does not imply association.

**Remark 1** (Two counterexamples). A positive mixture that is not associated. Let  $\Theta$  be fair on  $\{A, B\}$ , and given  $\Theta$  draw i.i.d. scores from  $P_A = \text{U}(1.4, 1.6)$  or  $P_B = \frac{1}{2}\text{U}(-0.1, 0.1) + \frac{1}{2}\text{U}(2.9, 3.1)$ . With the increasing functions  $f = \mathbf{1}\{s > 1\}$  and  $g = \mathbf{1}\{s > 2\}$ ,  $\text{Cov}(f(S_1), g(S_2)) = \frac{1}{8} - \frac{3}{4} \cdot \frac{1}{4} = -\frac{1}{16}$ . With  $n = 1$  the width is  $S_1$ , and a width near 0 or 3 points to  $B$  while a width near 1.5 points to  $A$ , so  $S_2$  is neither stochastically increasing nor decreasing in it.

A signed law that is positively correlated. Let  $(X_1, X_2, X_3)$  be exchangeable Bernoulli with  $\Pr(\sum X_i = 3) = \frac{1}{2}$  and  $\Pr(\sum X_i = 1) = \frac{1}{2}$ . Then  $\text{Cov}(X_i, X_j) = \frac{1}{2} - \frac{4}{9} = \frac{1}{18} > 0$ . A positive mixture would need moments  $\mathbb{E}P = \frac{2}{3}$  and  $\mathbb{E}P^2 = \mathbb{E}P^3 = \frac{1}{2}$ , but  $\mathbb{E}P^2(1-P) = 0$  forces  $P \in \{0, 1\}$  and hence  $\mathbb{E}P = \mathbb{E}P^2$ , a contradiction. So the law is not extendable.

### 3 Marginal coverage ignores the sign

For continuous exchangeable scores the rank of  $S_{n+1}$  among all  $n + 1$  is uniform, and  $S_{n+1} \leq S_{(k)}$  iff that rank is at most  $k$ , so

$$\mathbb{P}(S_{n+1} \leq \hat{q}) = \frac{k}{n+1} \geq 1 - \alpha. \quad (1)$$

This uses nothing but exchangeability (?). Positive or signed, the marginal number is the same. The sign enters only when we ask what the certificate hides.

### 4 The conditional-coverage decomposition

Condition on the realised width. For a value  $q$  of the width  $\hat{q}$  and a threshold  $s$ , write  $H(s, q) = \mathbb{P}(S_{n+1} \leq s \mid \hat{q} = q)$ . Coverage holds when  $S_{n+1} \leq \hat{q}$ , so coverage conditional on the realised width is the diagonal  $c(q) = H(q, q)$ , where the threshold equals the width.

**Lemma 1** (Conditional-coverage slope). *If  $(S_{n+1}, \hat{q})$  has a joint density, then*

$$c'(q) = \underbrace{f_{S_{n+1}|\hat{q}}(q \mid q)}_{\text{(i) threshold}} + \underbrace{\partial_q \mathbb{P}(S_{n+1} \leq q \mid \hat{q} = q)}_{\text{(ii) dependence}}. \quad (2)$$

*Term (i) is non-negative. Term (ii) is  $\leq 0$  when  $S_{n+1}$  is stochastically increasing in  $\hat{q}$ ,  $= 0$  under independence, and  $\geq 0$  when stochastically decreasing.*

*Proof.* Total derivative of  $H(q, q)$  along the diagonal:  $c'(q) = \partial_s H(s, q)|_{s=q} + \partial_q H(s, q)|_{s=q}$ . The first equals  $f_{S_{n+1}|\hat{q}}(q \mid q) \geq 0$ . For the second,  $\partial_q H(s, q) \leq 0$  for all  $s$  exactly when  $\mathbb{P}(S_{n+1} \leq s \mid \hat{q} = q)$  is non-increasing in  $q$ , the definition of  $S_{n+1}$  being stochastically increasing in  $\hat{q}$ . The reverse inequality is stochastic decrease.  $\square$

**Remark 2** (The independence case is classical). *Under independence term (ii) vanishes and  $c(q) = F(q)$ , the test-score CDF, which already rises with the width purely from estimation. This is the known law of calibration-conditional coverage:  $F(\hat{q}) = F(S_{(k)}) \sim \text{Beta}(k, n - k + 1)$  (???). Lemma 1 is its extension to dependent scores, with term (ii) the new piece.*

**Remark 3** (Reading the sign). *The direction is set by whether  $S_{n+1}$  is stochastically increasing or decreasing in  $\hat{q}$ , which is a property of the dependence, not of the sign of a de Finetti representation. A sufficient condition for the increasing case is a scalar latent location with log-concave noise,  $S_i = \Theta + \varepsilon_i$ . Order statistics of log-concave noise are log-concave, so the posterior of  $\Theta$  given  $\hat{q}$  rises in the likelihood-ratio order, and  $S_{n+1}$  rises stochastically with it. Term (ii) is then  $\leq 0$ , it offsets the non-negative threshold term, and pooling the calibration scores conditions you into the realised world. The decreasing case is the batch-relative one of Section 5 (without-replacement samples, multinomials, ranks, compositional vectors), where the calibration set anti-predicts the test and the per-case coverage swings high and low while the marginal stays pinned. A positive mixture without such monotone structure, as in Remark 1, can give term (ii) either sign at different widths.*

The reading concerns conformal's coverage, and coverage only. The residual-information gap of the companion paper sits there whatever the sign, so even at its best this is a coverage certificate, not a sharp forecast. When the test score rises with the width, the dependence term works against the threshold fan, and per-case coverage varies less with the width. When it falls, the two terms add, and it varies more: the number holds on average while the per-case coverage swings high and low. The sharpness comes from modelling the latent, never from the wrapper.

## 5 Why the signed corner is not exotic: constrained scores

The negative corner arises whenever the nonconformity score is defined relative to the batch, because a linear constraint forces it. De-meaning ( $\sum_i s_i = 0$ ), differencing, ranks or percentiles-within-batch, and compositional or budgeted targets all impose such a constraint, and the constraint is the negative association (?). For these scores conformal remains marginally valid (if the relativisation is symmetric across calibration and test, exchangeability is preserved), but its conditional coverage is the fanning, anti-adaptive kind of Lemma 1.

A caution: if the relativisation treats the test point asymmetrically (de-meaning by a training-only mean, say), exchangeability itself fails and even the marginal guarantee is no longer assured. This is the familiar reason split conformal uses out-of-fold rather than in-sample residuals (?); in-sample regression residuals are both constrained ( $\sum_i e_i = 0$ ) and heteroscedastic (variance  $1 - h_{ii}$ ), and so are not exchangeable to begin with.

**Remark 4** (Rank, choice, and market-implied scores). *The signed corner is the native habitat of contest data. In a Thurstone or Luce choice model a field of competitors carries latent values and the observed outcome is a winner or a full ranking. The winner indicator is a single multinomial trial, and the rank vector a permutation law, both negatively associated (?). Any nonconformity score read off such a field (a rank, a within-field relative score, or a market-implied win probability) therefore sits in the negatively-associated regime above, where the marginal certificate is least informative about conditional reliability. The converse direction is the modelling the certificate cannot supply: recovering latent means from observed prices or winning probabilities (?) is exactly the de Finetti latent-recovery of Section 2. In betting terms the market price is the crowd’s  $Q$ , the recovered latent mean is an estimate of the true  $P$ , and the divergence between them is the rent of the companion paper. So the same contest data that lands conformal in its anti-adaptive corner is also where the de Finetti latent, once estimated, pays.*

## 6 A runnable detector

The gap is defined through the log-score, so it is not something you read off data. Its energy-statistics counterpart is, and it gives a test you can run on any fitted conformal predictor.

**Lemma 2** (A distance-covariance detector). *Let  $U = \widehat{G}(R)$  be the conformal rank of the residual among the calibration residuals. Then  $U$  is marginally uniform, and  $\text{dCov}(U, X) = 0$  iff  $U \perp X$  iff  $R \perp X$  iff  $I(R; X) = 0$ . The permutation test that reshuffles the  $U$  labels against  $X$  is a valid test of  $R \perp X$ , and rejecting it certifies  $I(R; X) > 0$ : the conformal rank is conditionally non-uniform, so the marginal coverage is uneven across  $x$ .*

Two reasons this is the right object.  $U$  is standardized to uniform whatever the model, so the test probes only the conditional structure conformal cannot see. And distance covariance (?) catches dependence in any moment: under symmetric heteroscedastic noise the conditional mean of  $U$  stays at  $\frac{1}{2}$  while its spread moves with  $x$ , so a correlation test sees nothing and the distance covariance sees it.

A small experiment makes the point (one run, reproduced by `check_dcov.py`). The residual is  $R = \sigma(X)\varepsilon$  with the location correct, so any dependence is pure conditional shape.

| regime                  | marginal coverage | $\text{dCov}(U, X)$ | permutation $p$ | Pearson |
|-------------------------|-------------------|---------------------|-----------------|---------|
| homoscedastic           | 0.90              | 0.001               | 0.33            | +0.02   |
| heteroscedastic, mild   | 0.89              | 0.011               | 0.002           | +0.05   |
| heteroscedastic, strong | 0.90              | 0.023               | 0.002           | −0.03   |

Marginal coverage is on target throughout, so it cannot separate the regimes. The test fires only where the gap is positive, and the Pearson column shows that a correlation test would miss it.

The idea of a dependence measure as a conditional-coverage diagnostic is due to ?, who use HSIC between the miscoverage indicator and the interval length, and ?, who classify the indicator on  $x$ . Testing the rank against the features is the rank-based, level-free variant, and it is the energy counterpart of the gap:  $I(R; X)$  is the dependence of  $R$  and  $X$  in KL (?), and the distance covariance, equivalently HSIC (?), is the same dependence in energy. The paper’s gap is the one you prove; this is the one you run.

It detects, it does not certify. A large  $p$  means nothing was found at this power, not conditional validity, by the no-go results discussed in the companion paper (?). The ranks share the calibration ECDF, so leave-one-out or full-conformal ranks make the null exact. The test ships as `conformalguide.guardrails`.

## 7 Related work

The marginal-coverage fact is standard (?). The Beta law for coverage given the calibration set is ?, refined by ?; training-conditional coverage is ?. Magnitude-based bounds for dependent data are ? and ?. Association and negative association are ? and ?; finite and signed de Finetti are ? and ??.

? are closest: they show conformal can over- or under-cover under temporal dependence. That account is marginal, it grades a departure by the size of the dependence rather than its sign, and it does not condition on the calibration set. The sign of the dependence term, and conditions that fix it, are what this note adds. The sign of the finite de Finetti measure is related to it but does not determine it (Remark 1).

de Finetti describes a law. Conformal runs without one. Using a de Finetti measure means estimating it, and that estimation is ill-posed: the latent is recovered from finite data, so in general one regularizes, shrinking toward the field. Conformal skips this, and the residual-information gap is the price of skipping it. A signed measure gives nothing to estimate from at all, so the representation tells you how conformal behaves rather than standing in for it. ? argue conformal is not a de Finetti conditional at all.